

A Faithful One-Driver Complex Embedding of Arithmetic Brownian Motion

Itô dynamics, covariance rank, and zero crossings

Complex–Itô Random-Walk Research Project

July 23, 2026

We study how a scalar arithmetic Brownian motion can be represented by a complex exponential whose radius and angle are both stochastic without introducing a second Brownian driver. For $dX_t = \mu dt + \sigma dW_t$, $c = \alpha + i\beta$, and $Z_t = Z_0 \exp(c(X_t - X_0))$, the radius recovers the scalar state whenever $\alpha \neq 0$, while the unwrapped angle and log-radius lie on a deterministic line. We derive the induced complex Itô equation, exact finite-step multiplicative walk, distributions, moments, and both the bilinear and Hermitian complex quadratic variations. A general rank theorem shows that every smooth complex image of a scalar diffusion has instantaneous Cartesian covariance of rank at most one. This distinguishes the embedding from planar Brownian motion, whose local covariance has rank two. We also explain why the literal polar representation of a signed scalar process is only local before a zero and why Tanaka local time is required globally. Symbolic checks, exact-path identities, Monte Carlo estimates, shared-path strong-convergence experiments, and local-time approximations are preserved in an executable discovery notebook. The construction is an information-preserving geometric representation of one-dimensional noise, not a relabelling of planar Brownian motion.

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1 Introduction

The symbolic rule

$$(dW_t)^2 = dt$$

and the algebraic identity $i^2 = -1$ look tantalizingly similar. They are not the same kind of statement. The first abbreviates a quadratic-variation limit; the second is an exact identity in the complex numbers. Nevertheless, complex multiplication is valuable for organizing stochastic scale and rotation. The question is how to use that geometry without changing the underlying stochastic model.

Let

$$dX_t = \mu dt + \sigma dW_t, \quad X_0 \in \mathbb{R}, \quad (1)$$

where W is one standard real Brownian motion. We seek a complex process whose radius and angle both fluctuate, which contains exactly the information in X , and which is still driven by this one W . The central construction is

$$\boxed{Z_t = Z_0 \exp(c(X_t - X_0)), \quad c = \alpha + i\beta, \quad \alpha \neq 0.} \quad (2)$$

Its polar coordinates are

$$\log R_t = \log R_0 + \alpha(X_t - X_0), \quad \Theta_t = \Theta_0 + \beta(X_t - X_0). \quad (3)$$

Both coordinates are stochastic if $\alpha\beta \neq 0$, but they share one Brownian direction. The resulting path lies on a logarithmic spiral and has rank-one instantaneous covariance. That rank is the mathematical distinction between a faithful complex embedding of scalar Brownian motion and a genuine two-driver Brownian motion in the plane.

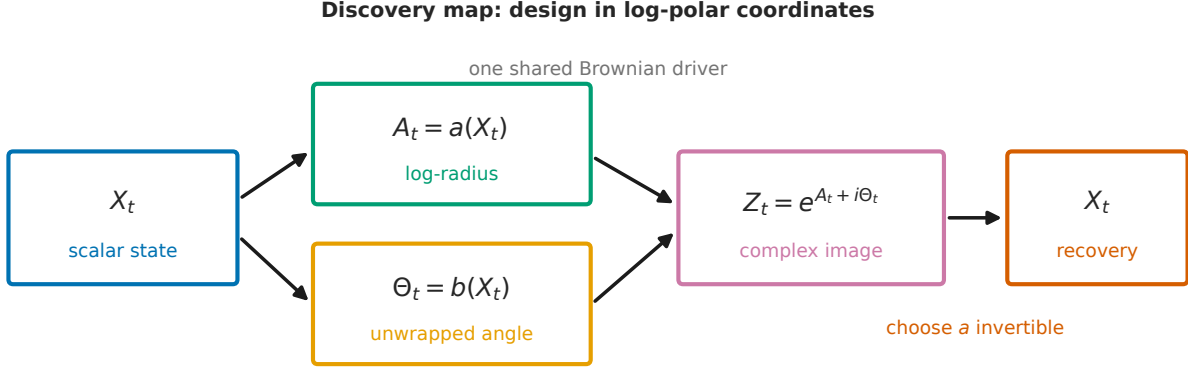


Figure 1: Starting from a scalar state, design log-radius and unwrapped angle as functions of that state, reconstruct a complex variable, and recover the state through an invertible radius map.

Figure 1 summarizes the constructive strategy. The contribution is a rigorous synthesis and visual account of standard Itô tools applied to this specific design problem. We do **not** claim that the exponential map, the logarithmic spiral, or the outer-product rank calculation is a new theorem.

The main results are:

1. an injective one-driver complex embedding with stochastic radius and angle;
2. two matching derivations of its complex Itô equation;
3. exact state recovery and an exact finite-step multiplicative random walk;
4. a general rank-one obstruction for smooth complex images of scalar diffusions;
5. explicit complex brackets, distributions, and moments; and
6. a localized and global treatment of scalar zero crossings.

2 Literature and positioning

The multidimensional Itô formula and the covariance factorization $\Sigma\Sigma^\top$ are standard. We use them componentwise for complex maps, treating $\mathbb{C}$ as $\mathbb{R}^2$; accessible derivations appear in Goodman’s stochastic-calculus notes and the broad survey by Pitman and Yor (Goodman 2007; Pitman and Yor 2018).

The distinction between stochastic exponentials and stochastic logarithms is important near zeros. Larsson and Ruf formulate these transforms on stochastic intervals and make explicit why continuous hitting of zero matters (Larsson and Ruf 2019). Brownian local time, occupation density, and Tanaka’s formula are standard tools for the nonsmooth absolute-value map (Mörters and Peres 2010; Biskup 2025).

Planar Brownian motion supplies a necessary control, not an alternative description of the scalar model. Its radial Bessel behavior, nonhitting of a fixed point when started elsewhere, and winding process belong to a genuinely two-dimensional theory (Pitman and Yor 2018, 1986). The numerical strong-convergence experiment follows the shared-Brownian-path methodology used in expositions of Euler–Maruyama (Higham 2001).

The particular synthesis developed below—designing a recoverable log-polar embedding, stating its noise-rank obstruction, and relating it to exact complex random-walk steps—is derived directly from these standard tools. The paper is therefore best read as an expository computational mathematics note.

3 Preliminaries and notation

We identify $z = x + iy \in \mathbb{C}$ with $(x, y) \in \mathbb{R}^2$. For a complex continuous semimartingale $Z = U + iV$, define

$$[Z, Z] = [U, U] - [V, V] + 2i[U, V] \quad (4)$$

and

$$[Z, \overline{Z}] = [U, U] + [V, V]. \quad (5)$$

The first is complex bilinear and can exhibit phase cancellation. The second is Hermitian, real, and nonnegative.

The angle Θ in this paper is an *unwrapped* continuous lift, not merely an element of $\mathbb{R}/(2\pi\mathbb{Z})$. This distinction matters when discussing recovery from phase.

Table 1: Core notation.

Symbol	Meaning
X_t	scalar arithmetic Brownian motion
W_t	one standard real Brownian driver
μ, σ	scalar drift and volatility, with $\sigma > 0$
Z_t	complex embedding
$R_t = Z_t $	stochastic radius
Θ_t	unwrapped stochastic angle
$A_t = \log R_t$	log-radius
$c = \alpha + i\beta$	complex embedding parameter
$[Y_1, Y_2]_t$	quadratic covariation
τ_0	first hitting time of zero by a scalar process
$L_t^0(X)$	symmetric local time of X at zero

Three levels of equality will be kept separate:

- a **pathwise identity**, such as Eq. 2;
- an **Itô differential identity**, determined by finite variation and quadratic covariation; and
- a **mnemonic**, such as $(dW)^2 = dt$, which must not be read as an ordinary finite-increment algebraic equation.

4 Problem formulation

The word *representation* can mean several different things.

Table 2: Four meanings of representation.

Meaning	Requirement	What it preserves
Deterministic image	$Z_t = F(X_t)$	a chosen geometric view
State-injective embedding	F is injective	pointwise scalar state
Pathwise recovery	the lifted Z -path determines X	path information
Planar-model equality	the local law and covariance match a target in $\mathbb{R}^2$	full planar stochastic model

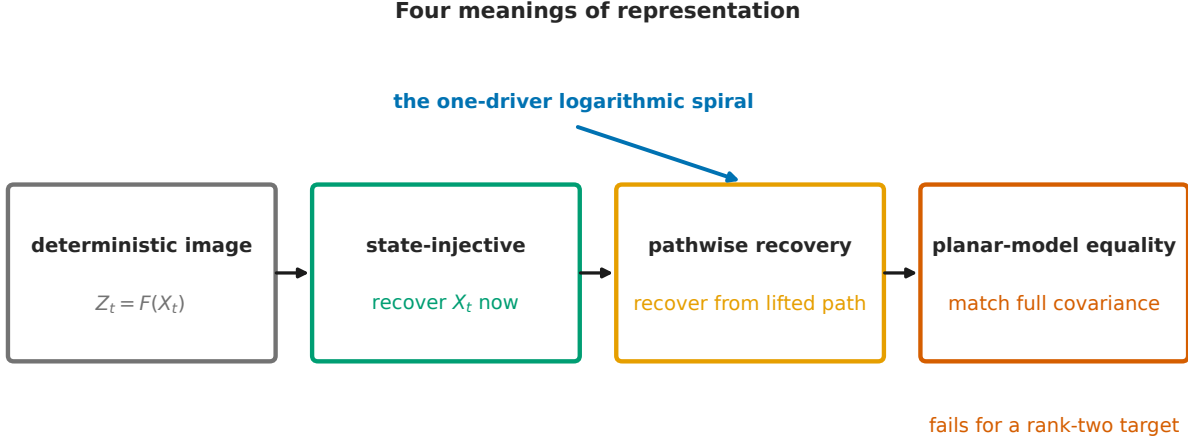


Figure 2: The logarithmic spiral satisfies deterministic image, state-injective embedding, and pathwise recovery, but it cannot equal a full-rank planar Brownian model.

The desired object must satisfy the first three notions while retaining one Brownian driver. The fourth is neither required nor possible for a rank-two target. Figure 2 makes this hierarchy explicit.

Write general log-polar coordinates as

$$A_t = a(X_t), \quad \Theta_t = b(X_t), \quad Z_t = e^{A_t + i\Theta_t}. \quad (6)$$

For C^2 functions a, b , scalar Itô calculus already guarantees that the two coordinates share one driver:

$$\begin{aligned} dA_t &= (\mu a'(X_t) + \tfrac{1}{2}\sigma^2 a''(X_t)) dt + \sigma a'(X_t) dW_t, \\ d\Theta_t &= (\mu b'(X_t) + \tfrac{1}{2}\sigma^2 b''(X_t)) dt + \sigma b'(X_t) dW_t. \end{aligned} \quad (7)$$

Thus the design problem becomes deterministic: choose a invertible for state recovery and choose both derivatives nonzero for stochastic radius and angle.

5 The logarithmic-spiral construction

The simplest globally smooth choice is affine:

$$a(x) = \log R_0 + \alpha(x - X_0), \quad b(x) = \Theta_0 + \beta(x - X_0). \quad (8)$$

Substitution in Eq. 6 gives Eq. 2.

Theorem 5.1 (Faithful one-driver embedding). *Let X satisfy Eq. 1, let $Z_0 \neq 0$, and let $c = \alpha + i\beta$ with $\alpha \neq 0$. Then the process in Eq. 2 is nonvanishing and state-injective. Specifically,*

$$X_t = X_0 + \frac{1}{\alpha} \log \frac{|Z_t|}{|Z_0|}. \quad (9)$$

Consequently, after completion in the usual way, $\sigma(Z_s : 0 \leq s \leq t) = \sigma(X_s : 0 \leq s \leq t)$ for each t .

The proof is immediate from $|Z_t| = R_0 e^{\alpha(X_t - X_0)}$. The formula also shows why the lifted angle is not needed for recovery when $\alpha \neq 0$.

Eliminating $X_t - X_0$ from Eq. 3 gives

$$\boxed{\Theta_t - \Theta_0 = \frac{\beta}{\alpha} \log \frac{R_t}{R_0}.} \quad (10)$$

Thus the two visible polar coordinates are exactly coupled.

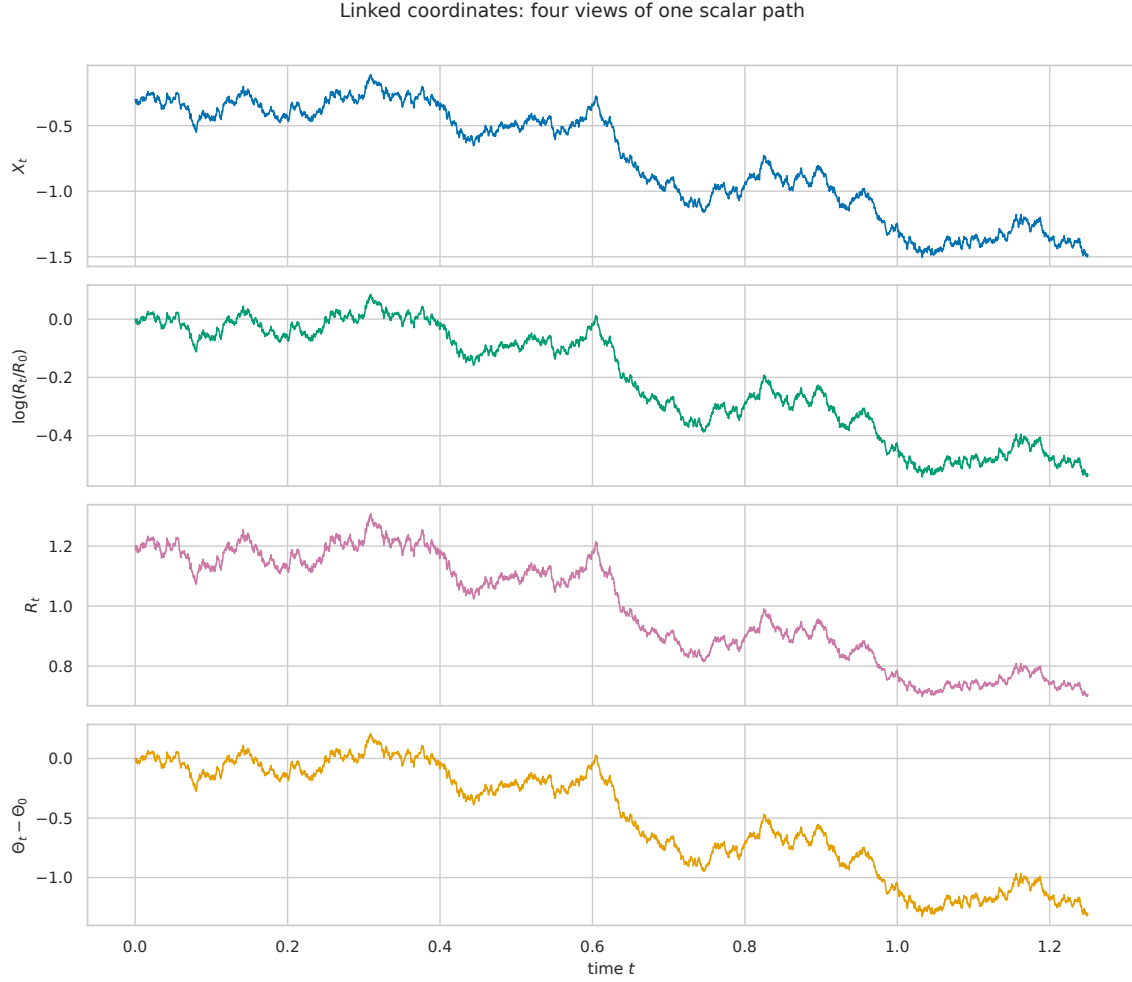


Figure 3: The scalar path, log-radius, radius, and unwrapped angle share every reversal because they are deterministic functions of one state.

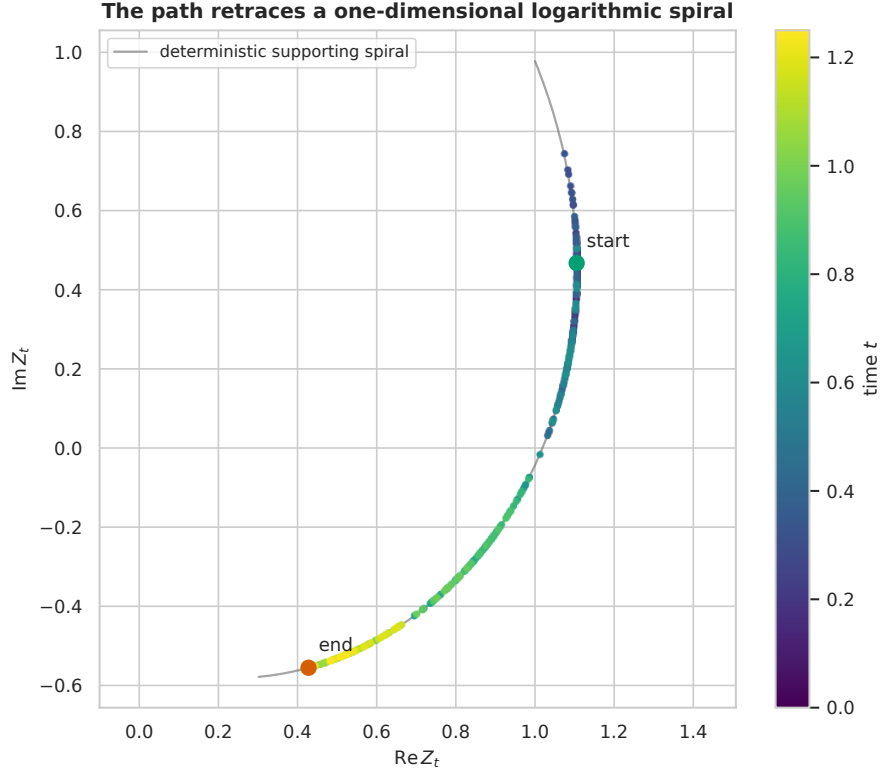


Figure 4: A time-colored sample path lies on and repeatedly retraces one deterministic logarithmic spiral.

The parameter cases are:

Table 3: Parameter classification.

Parameters	Geometry	Stochastic information
$\alpha = 0, \beta \neq 0$	circle	radius loses the state; lifted phase can retain it
$\alpha \neq 0, \beta = 0$	ray	radius alone is injective
$\alpha\beta \neq 0$	logarithmic spiral	radius and angle fluctuate, rank one
$\alpha = \beta = 0$	one point	all state information is lost

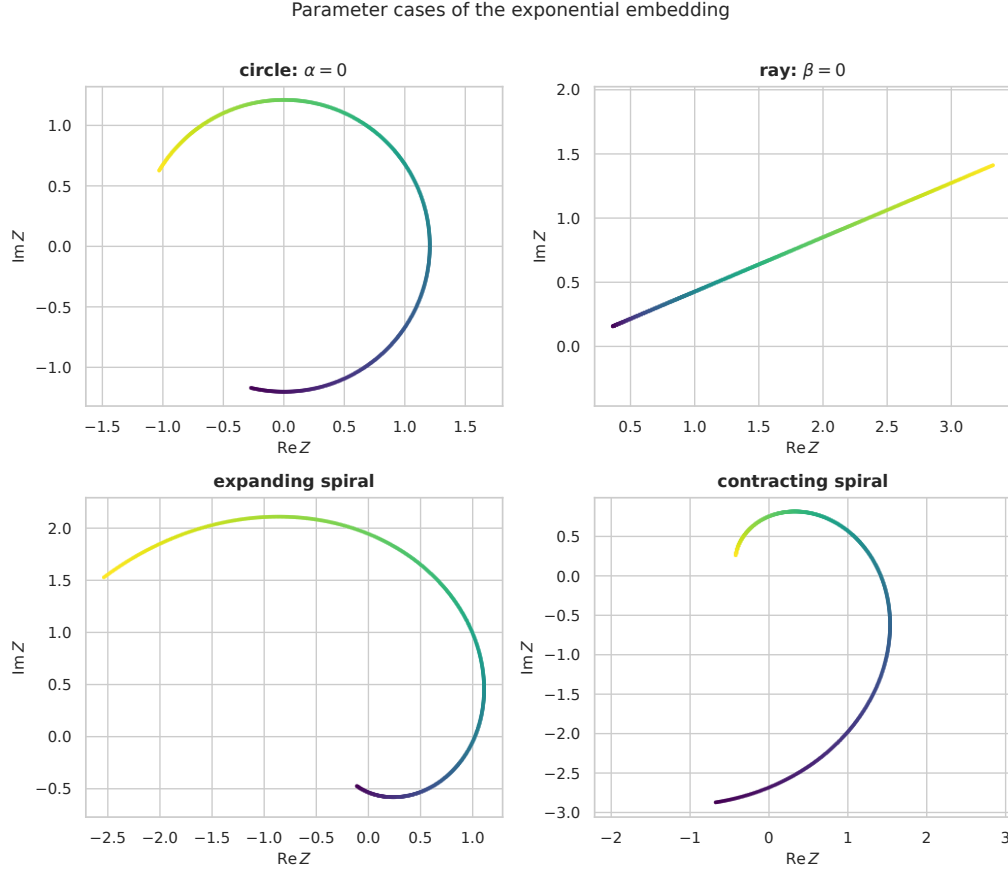


Figure 5: Changing alpha and beta produces a circle, a ray, an expanding spiral, or a contracting spiral.

6 Itô dynamics and the exact random walk

Let $F(x) = Z_0 e^{c(x-X_0)}$. Since $F' = cF$ and $F'' = c^2 F$, scalar Itô calculus gives

$$\boxed{\frac{dZ_t}{Z_t} = \left(c\mu + \frac{1}{2}c^2\sigma^2 \right) dt + c\sigma dW_t.} \quad (11)$$

Equivalently,

$$dZ_t = \lambda Z_t dt + \nu Z_t dW_t, \quad \lambda = c\mu + \frac{1}{2}c^2\sigma^2, \quad \nu = c\sigma. \quad (12)$$

The differential dZ_t/Z_t in Eq. 11 is the stochastic logarithm differential of Z (Larsson and Ruf 2019).

There is an independent polar derivation. From Eq. 3,

$$dA_t = \alpha\mu dt + \alpha\sigma dW_t, \quad d\Theta_t = \beta\mu dt + \beta\sigma dW_t. \quad (13)$$

For $Z = e^{A+i\Theta}$, two-dimensional Itô calculus yields

$$\frac{dZ}{Z} = dA + i d\Theta + \frac{1}{2} (d[A, A] - d[\Theta, \Theta]) + i d[A, \Theta]. \quad (14)$$

Using

$$d[A, A] = \alpha^2 \sigma^2 dt, \quad d[\Theta, \Theta] = \beta^2 \sigma^2 dt, \quad d[A, \Theta] = \alpha \beta \sigma^2 dt \quad (15)$$

reduces Eq. 14 exactly to Eq. 11. The mixed covariation is essential; omitting it loses the imaginary Itô correction.

Because Eq. 2 is pathwise, a grid increment has the exact update

$$\boxed{Z_{n+1} = Z_n \exp(c \Delta X_n), \quad \Delta X_n = \mu \Delta t + \sigma \Delta W_n.} \quad (16)$$

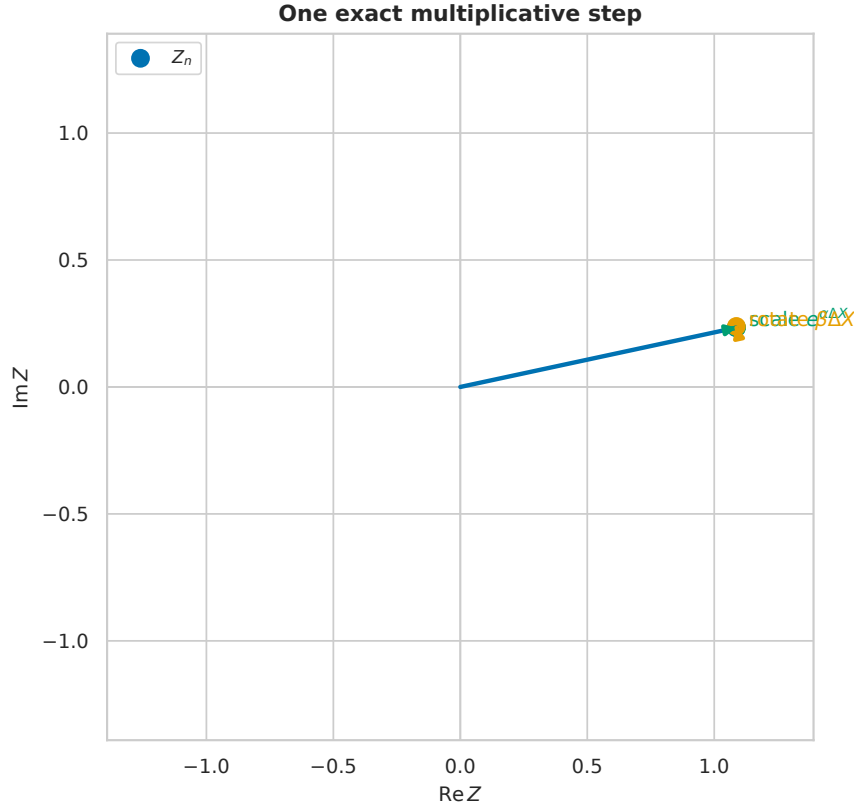


Figure 6: One scalar increment first appears as radial scaling and then as angular rotation; together they equal the exact complex factor.

Eq. 16 is not Euler–Maruyama. It is exact at every grid point because the exponential factors telescope.

The diffusion coefficient in Eq. 11 is $c\sigma Z_t$, a single complex vector. It determines one instantaneous direction in $\mathbb{R}^2$, even though the direction rotates with the state.

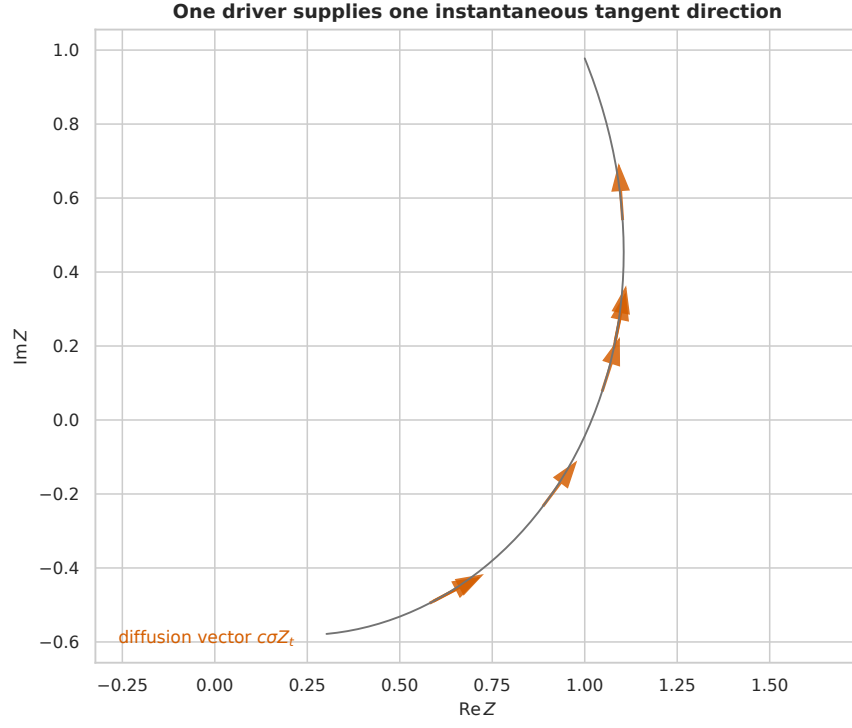


Figure 7: Red arrows show the single diffusion direction tangent to the supporting spiral at several states.

The complex brackets follow immediately from the diffusion term:

$$\boxed{d[Z, Z]_t = c^2 \sigma^2 Z_t^2 dt, \quad d[Z, \bar{Z}]_t = |c|^2 \sigma^2 |Z_t|^2 dt.} \quad (17)$$

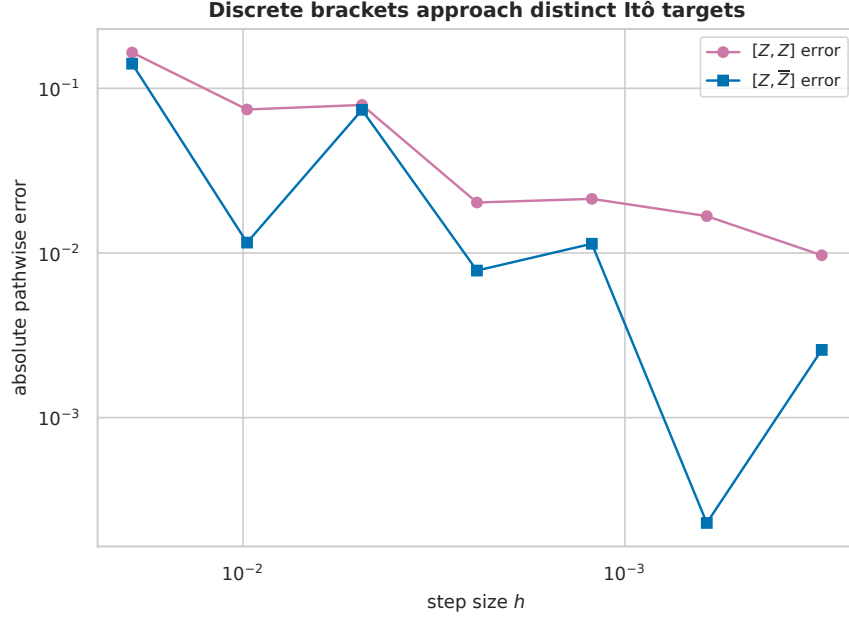


Figure 8: Discrete bilinear and Hermitian bracket sums approach their different Itô-integral targets as the grid is refined.

The pathwise convergence in Figure 8 is irregular, as expected for one sample path, but both finest-grid errors are substantially below their coarsest-grid values.

Finally, $Y_t = X_t - X_0 \sim N(\mu t, \sigma^2 t)$, so

$$\mathbb{E}[R_t^p] = R_0^p \exp(p\alpha\mu t + \frac{1}{2}p^2\alpha^2\sigma^2 t) \quad (18)$$

and, for integers n ,

$$\mathbb{E}\left[\left(\frac{Z_t}{Z_0}\right)^n\right] = \exp(nc\mu t + \frac{1}{2}n^2c^2\sigma^2 t). \quad (19)$$

7 Noise dimension and covariance rank

The log-polar covariance density from Eq. 13 is

$$K = \sigma^2 \begin{pmatrix} \alpha^2 & \alpha\beta \\ \alpha\beta & \beta^2 \end{pmatrix} = \sigma^2 \begin{pmatrix} \alpha \\ \beta \end{pmatrix} (\alpha \quad \beta). \quad (20)$$

Its eigenvalues are 0 and $\sigma^2(\alpha^2 + \beta^2)$. Both coordinates fluctuate, but their local correlation has absolute value one when $\alpha\beta \neq 0$.

Theorem 7.1 (Rank-one theorem for scalar complex images). *Let*

$$dX_t = b_t dt + s_t dW_t$$

be a scalar Itô process and let $F = (F_1, F_2) : \mathbb{R} \rightarrow \mathbb{R}^2$ be C^2 . Then the instantaneous Cartesian covariance density of $F(X_t)$ is

$$C_t = s_t^2 F'(X_t) F'(X_t)^\top, \quad (21)$$

and therefore

$$\text{rank } C_t \leq 1. \quad (22)$$

Proof. Componentwise Itô calculus gives

$$dF_j(X_t) = (b_t F'_j(X_t) + \frac{1}{2} s_t^2 F''_j(X_t)) dt + s_t F'_j(X_t) dW_t.$$

The two-dimensional diffusion matrix has the single column $s_t F'(X_t)$. Its covariance is the outer product in Eq. 21, whose rank is at most one. $\square$

The same conclusion holds for any adapted two-dimensional Itô process driven by one scalar Brownian motion:

$$dY_t = q_t dt + v_t dW_t \implies \frac{d[Y, Y]_t}{dt} = v_t v_t^\top. \quad (23)$$

Path dependence and state dependence can rotate v_t , but cannot create a second simultaneous noise direction.

For comparison only, isotropic planar Brownian motion

$$d\tilde{Z}_t = \sigma(dW_t^1 + i dW_t^2) \quad (24)$$

has Cartesian covariance $\sigma^2 I_2$, of rank two. At a fixed nonzero radius r , its log-polar covariance is $(\sigma^2/r^2) I_2$. The numerical control below uses $r = 1.30$, so its theoretical log-polar eigenvalues are $\sigma^2/r^2 = 0.4225/1.69 = 0.25$. This is a distinct stochastic model with two independent Brownian drivers (Pitman and Yor 2018).

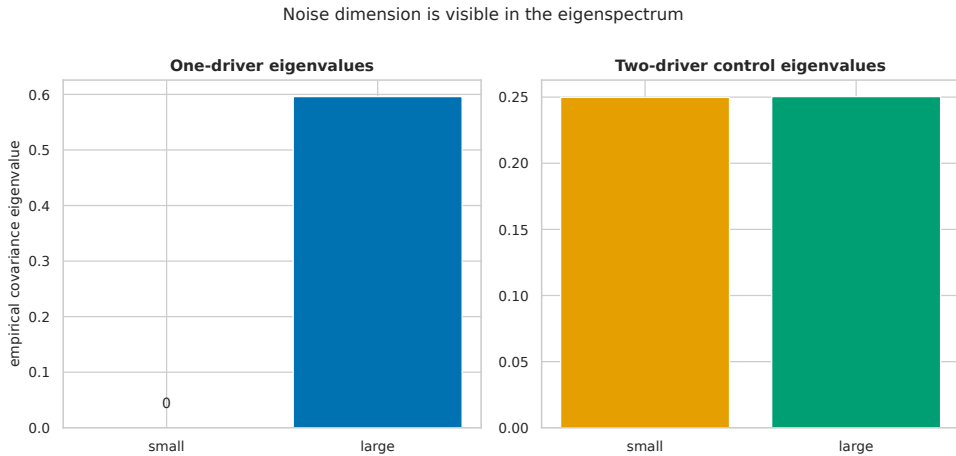
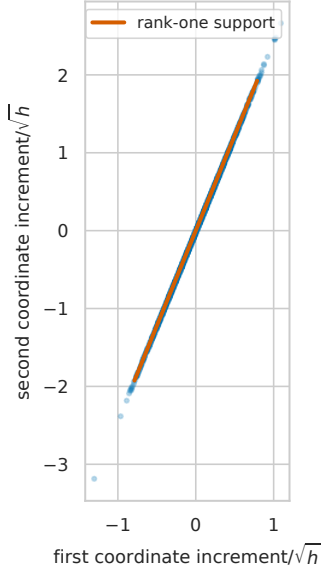


Figure 9: The empirical one-driver covariance has one numerical-zero eigenvalue, while the two-driver control has two positive eigenvalues.

One driver: collapsed covariance ellipse



Two drivers: full-rank covariance ellipse

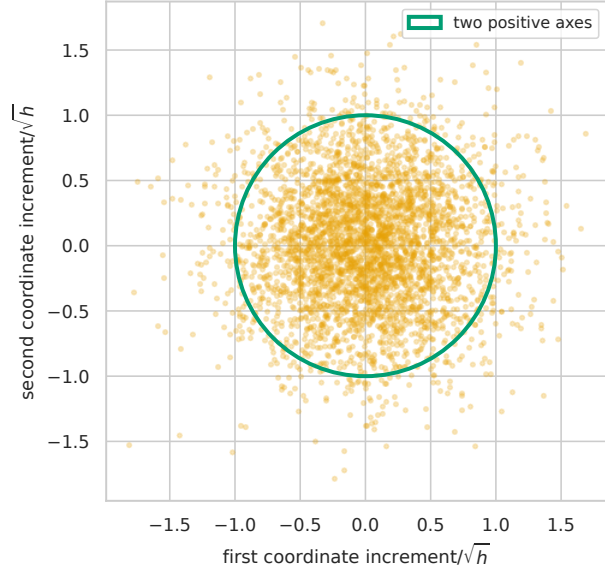


Figure 10: Normalized one-driver increments collapse to a line, while two-driver control increments occupy a two-dimensional cloud.

This distinction concerns the local covariance, not the long-time visual support alone. A one-driver diffusion can trace a curved subset of the plane, and a state-dependent one-driver diffusion can even have two-dimensional support over time. At each instant, however, its continuous martingale increment still lies in one direction.

8 Scalar polar coordinates and zero crossings

A literal embedding of the signed scalar state is

$$Z_t^{\text{axis}} = X_t + i0. \quad (25)$$

Away from zero, $R_t = |X_t|$ and $\Theta_t = 0$ for $X_t > 0$, $\Theta_t = \pi$ for $X_t < 0$. Let

$$\tau_0 = \inf\{t \geq 0 : X_t = 0\}.$$

On $t < \tau_0$, scalar Itô calculus gives the localized identity

$$d \log |X_t| = \left(\frac{\mu}{X_t} - \frac{\sigma^2}{2X_t^2} \right) dt + \frac{\sigma}{X_t} dW_t. \quad (26)$$

This is exact on the stochastic interval before the zero, consistent with the general need to stop logarithms at continuous zeros (Larsson and Ruf 2019). It is not a global formula through sign changes.

The correct global formula for the radius is Tanaka's formula:

$$|X_t| = |X_0| + \int_0^t \text{sgn}(X_s) dX_s + L_t^0(X). \quad (27)$$

Here $L_t^0(X)$ is symmetric local time at zero (Mörters and Peres 2010; Biskup 2025). With constant volatility,

$$L_t^0(X) = \lim_{\varepsilon \downarrow 0} \frac{1}{2\varepsilon} \int_0^t \mathbf{1}_{\{|X_s| < \varepsilon\}} d[X, X]_s = \lim_{\varepsilon \downarrow 0} \frac{\sigma^2}{2\varepsilon} \int_0^t \mathbf{1}_{\{|X_s| < \varepsilon\}} ds \quad (28)$$

in the standard sense.

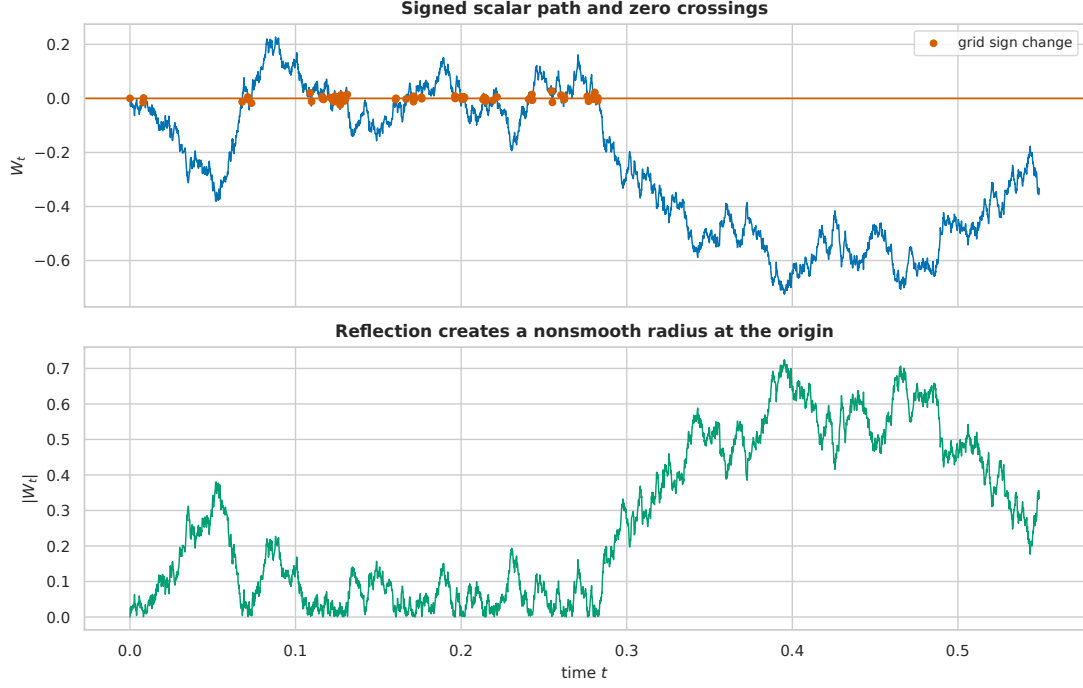


Figure 11: A signed Brownian path crosses zero while its absolute value reflects, creating the nonsmooth point that invalidates a global ordinary logarithm.

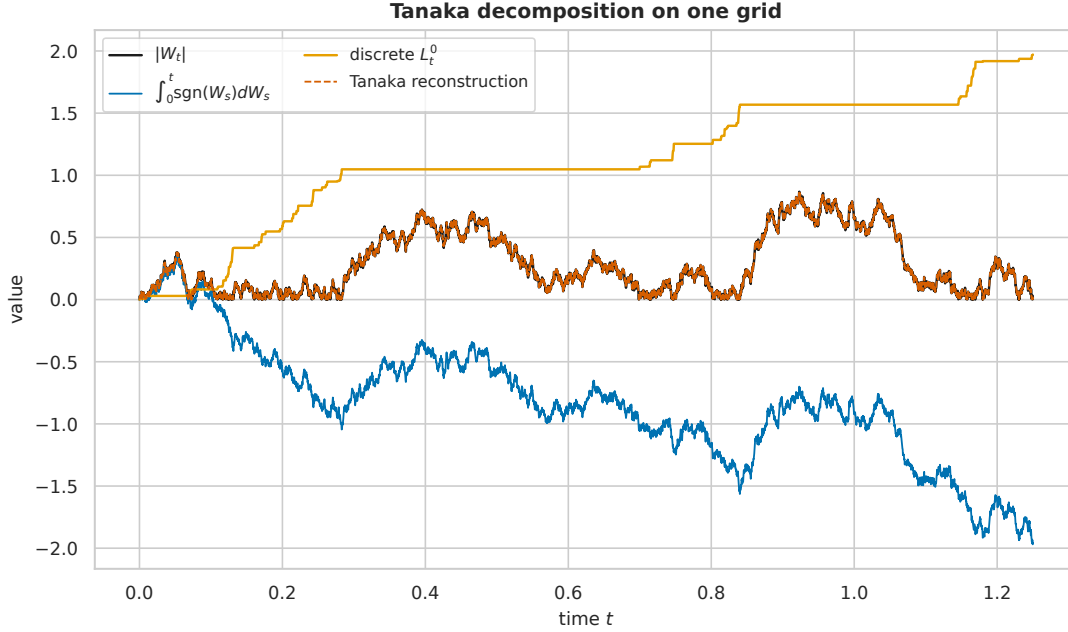


Figure 12: The reflected path equals the stochastic sign integral plus the accumulated discrete local-time residual.

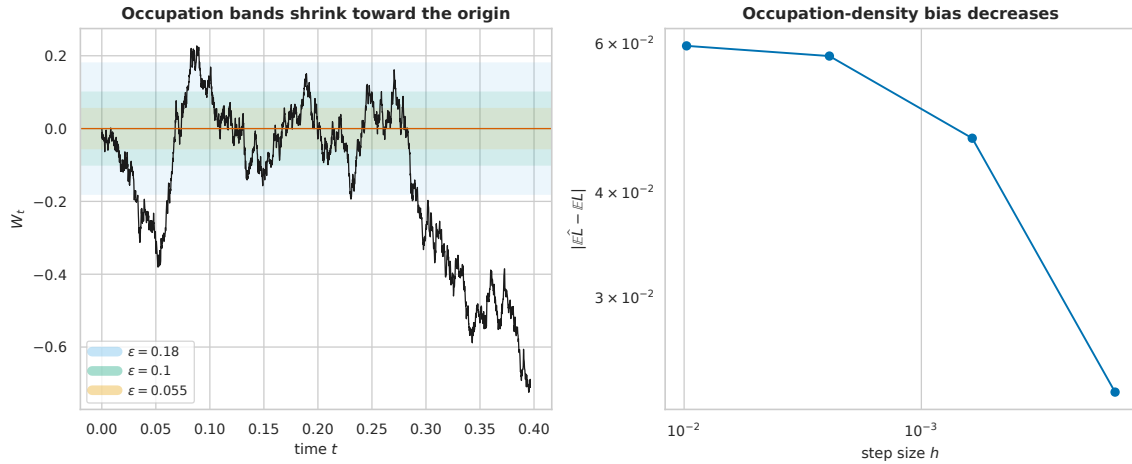


Figure 13: Shrinking occupation bands concentrate around zero, and the occupation-density bias decreases under joint grid and bandwidth refinement.

The logarithmic-spiral embedding avoids this singularity because $|Z_t| = R_0 e^{\alpha(X_t - X_0)} > 0$ for every finite X_t . It should not be confused with the literal polar coordinates of $X_t + i0$; the two representations answer different geometric questions.

9 Numerical methods

The executable supplement is `notebooks/phase3_discovery.ipynb`. It was generated by `notebooks/build_phase3_discovery` and executed from a clean kernel in the `phase3-paper` conda environment.

The principal parameters were

$$\mu = 0.20, \quad \sigma = 0.65, \quad \alpha = 0.45, \quad \beta = 1.10, \quad X_0 = -0.30, \quad R_0 = 1.20, \quad \Theta_0 = 0.40, \quad T = 1.25. \quad (29)$$

The master seed was 20260723; independent experiments used deterministic offsets from that seed.

The experiments were:

1. exact Gaussian simulation of a 4096-step arithmetic Brownian path;
2. pathwise comparison of the direct map, cumulative exact products, inverse radius, and spiral constraint;
3. 200,000 independent terminal draws for radial and complex moments;
4. 250,000 local increments for rank-one and rank-two covariance estimates;
5. 10,000 Brownian paths coupled across six Euler–Maruyama resolutions;
6. discrete bilinear and Hermitian bracket sums on seven nested grids;
7. a pathwise discrete Tanaka identity; and
8. 6,000-path local-time estimates on four grids.

For the exact discrete Tanaka check we used

$$\ell_n = |W_{n+1}| - |W_n| - \text{sgn}(W_n)\Delta W_n. \quad (30)$$

The sum of ℓ_n closes the discrete identity algebraically. The separate occupation estimator used

$$\hat{L}_{T,\varepsilon,h}^0 = \frac{1}{2\varepsilon} \sum_n \mathbf{1}_{\{|W_n| < \varepsilon\}} h, \quad \varepsilon = 0.5h^{1/4}. \quad (31)$$

The analytic derivations are the proofs. Monte Carlo estimates test the code, scale of uncertainty, and visual consequences.

10 Numerical results

10.1 Pathwise identities

The maximum errors along the 4096-step path were:

Table 4: Pathwise exactness checks; the residuals are floating-point roundoff.

Identity	Maximum absolute error
cumulative exact product versus direct Z_t	6.273×10^{-15}
X_t versus inverse-radius recovery	1.110×10^{-15}
logarithmic-spiral constraint	5.274×10^{-16}

The space-time view in Figure 14 is intentionally three-dimensional only to display temporal ordering. It must not be read as evidence of a second planar noise direction.

Space-time trajectory: planar position does not imply planar noise

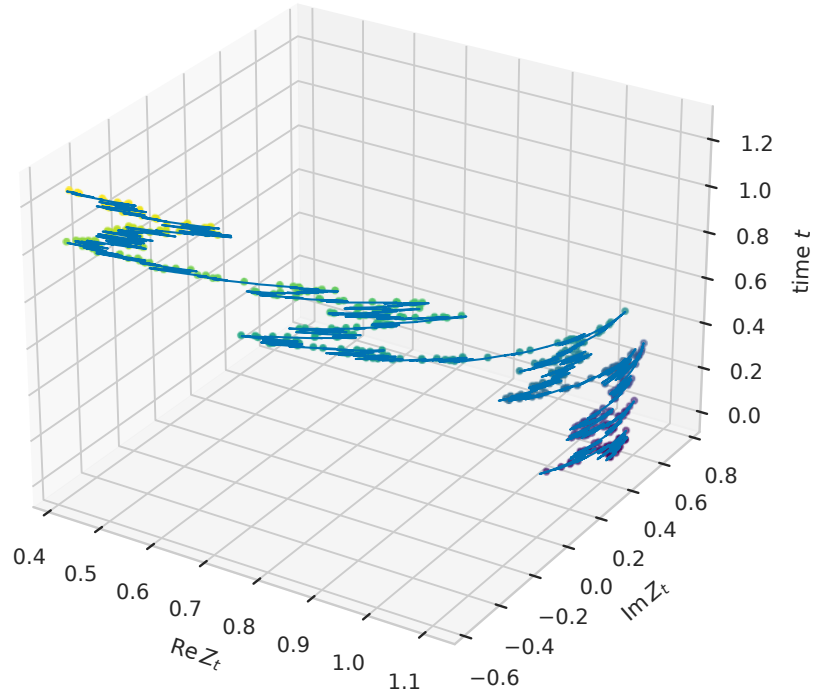


Figure 14: Adding time as the third axis separates repeated visits along the spiral without changing the one-driver interpretation.

10.2 Distribution and moment checks

The radial samples agree with the lognormal law implied by Eq. 18.

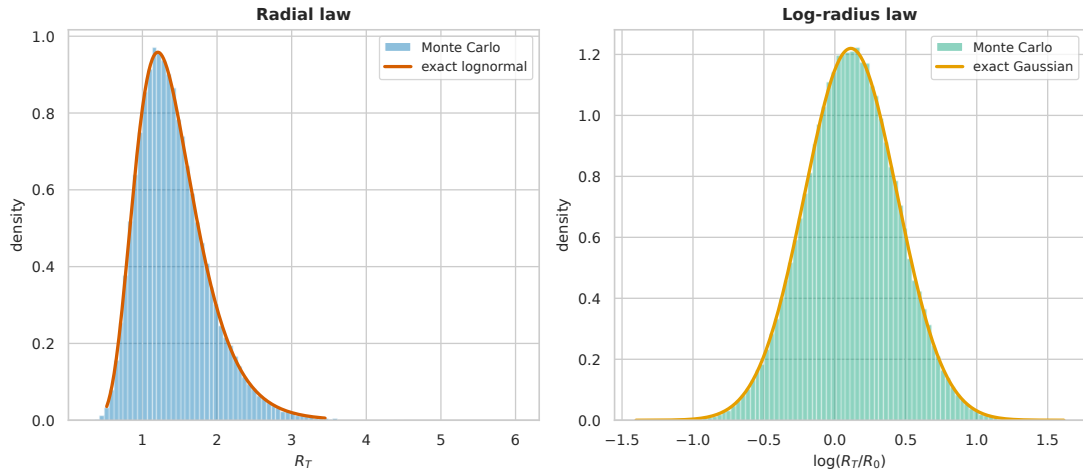


Figure 15: Monte Carlo histograms of radius and log-radius overlay their exact lognormal and Gaussian densities.

Table 5: Monte Carlo moment validation with 200,000 terminal draws.

Quantity	Component	Estimate	Exact	Standard error	z-score
$\mathbb{E}[R_T]$	real	1.415923	1.416649	0.001063	-0.683
$\mathbb{E}[R_T^2]$	real	2.230873	2.233419	0.003636	-0.700
$\mathbb{E}[Z_T/Z_0]$	real	0.737468	0.737199	0.001063	0.253
$\mathbb{E}[Z_T/Z_0]$	imaginary	0.437494	0.438321	0.001715	-0.482
$\mathbb{E}[(Z_T/Z_0)^2]$	real	-0.009798	-0.010754	0.003038	0.315
$\mathbb{E}[(Z_T/Z_0)^2]$	imaginary	0.431580	0.431934	0.002867	-0.124

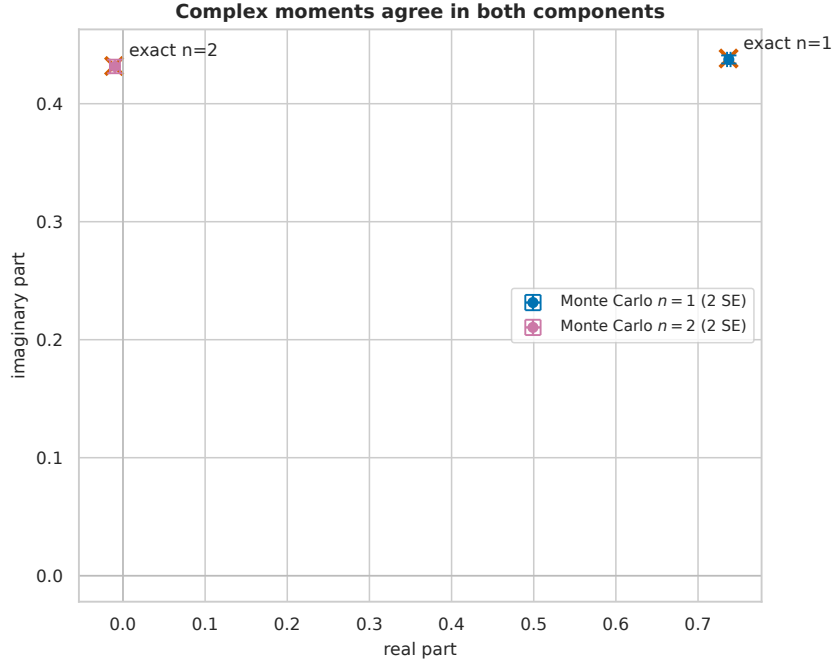


Figure 16: Empirical complex moments with two-standard-error bars coincide with their exact targets in the complex plane.

10.3 Covariance rank

Table 6: Empirical log-polar covariance eigenspectra. The tiny negative value is roundoff from a theoretically positive-semidefinite rank-one matrix.

Model	Small eigenvalue	Large eigenvalue	Relative matrix error
one-driver log-polar embedding	-1.39×10^{-16}	0.595978	0.00135
two-driver log-polar control ($r = 1.30$)	0.249766	0.250352	0.00120

These values numerically expose the exact rank distinction already proved in Theorem 7.1. The experiment is a check of implementation, not an independent proof.

10.4 Strong convergence

Table 7: Coupled Euler–Maruyama strong errors.

Steps N	h	Mean terminal error	Standard error
16	0.078125	0.150392	0.001469
32	0.0390625	0.106417	0.000972
64	0.0195313	0.075794	0.000690
128	0.00976563	0.052824	0.000455
256	0.00488281	0.037456	0.000311
512	0.00244141	0.026537	0.000229

The fitted log-log slope was 0.5015, consistent with strong order 1/2 (Higham 2001).

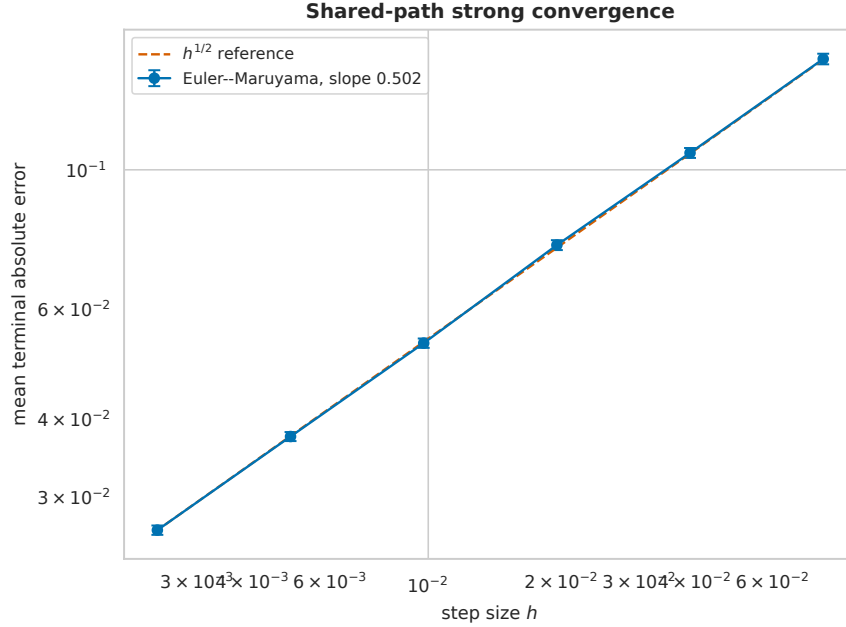


Figure 17: The shared-path Euler–Maruyama errors follow the half-order reference line.

10.5 Local time

For standard Brownian motion over $T = 1.25$,

$$\mathbb{E}[L_T^0] = \sqrt{\frac{2T}{\pi}} = 0.892062.$$

Table 8: Discrete Tanaka and occupation-density results over 6,000 paths.

Steps N	ϵ	Discrete mean	Occupation mean	Occupation absolute error	Identity max error
128	0.157179	0.895332	0.832549	0.059513	2.82×10^{-15}

Steps N	ϵ	Discrete mean	Occupation mean	Occupation absolute error	Identity max error
512	0.111142	0.882717	0.834180	0.057882	7.11×10^{-15}
2048	0.078590	0.884203	0.845796	0.046266	1.07×10^{-14}
8192	0.055571	0.893896	0.868905	0.023157	2.00×10^{-14}

The discrete identity closes to roundoff on every grid. The occupation estimator has visible finite-bandwidth bias, which decreases but converges more slowly. This is why exact grid hits are not used as a local-time estimator.

11 Discussion

The construction resolves the apparent tension between a stochastic radius and a scalar Brownian driver. A random radius does not imply two-dimensional noise. In Eq. 3, both log-radius and angle are random because both depend on X_t . They are nevertheless perfectly locally coupled, as Eq. 10 and Eq. 20 show.

The embedding preserves more than a visual trace. For $\alpha \neq 0$, Eq. 9 reconstructs the scalar state at every time, so the complex and scalar processes generate the same information. Yet it preserves less than a full planar model: no deterministic or adapted transformation driven by one scalar Brownian motion can match a positive-definite instantaneous two-dimensional covariance.

This distinction clarifies the roles of i and quadratic variation:

- multiplication by i rotates a complex vector by 90° ;
- the imaginary part of c converts a scalar displacement into phase;
- $i^2 = -1$ is exact complex algebra;
- $[W, W]_t = t$ is a pathwise limit of squared increments; and
- the Itô drift correction in Eq. 11 comes from the second derivative and quadratic variation, not from treating dW as an ordinary square root of dt .

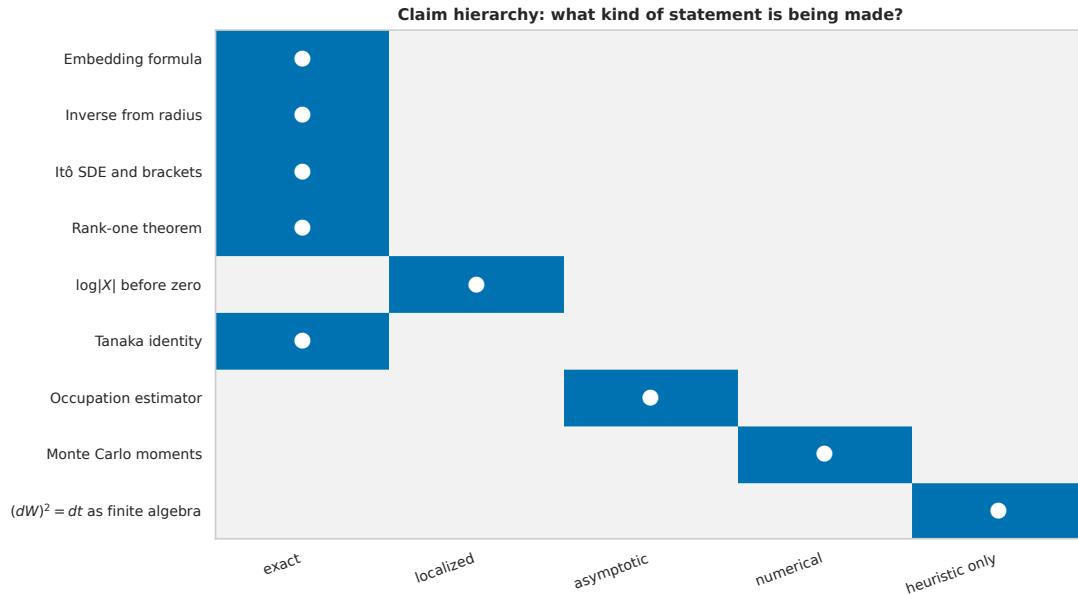


Figure 18: The notebook classifies each major statement as exact, localized, asymptotic, numerical, or heuristic.

Figure 18 is a compact guard against overinterpretation. In particular, the finite-increment statement $(\Delta W)^2 = \Delta t$ is false pathwise. Only aggregated squared increments converge to elapsed time.

12 Limitations

The construction has clear boundaries.

1. It is a deterministic embedding of a scalar process, not a universal model of planar stochastic motion.
2. When $\alpha = 0$, radius no longer provides pointwise recovery. A continuously lifted phase can preserve the scalar path, but the wrapped complex state need not be injective.
3. The literal real-axis polar form is singular at zero; localization and Tanaka local time are structural necessities, not numerical nuisances.
4. The covariance theorem concerns instantaneous continuous-martingale rank. It does not assert that every one-driver process has one-dimensional long-time support.
5. The occupation-density estimator converges slowly because grid and bandwidth limits interact.
6. Monte Carlo results validate implementation and scale; they do not prove the analytic identities.
7. No claim of historical novelty is made for the logarithmic-spiral map or the rank-one outer-product argument.

13 Conclusion

A scalar arithmetic Brownian motion has the faithful complex form

$$Z_t = Z_0 e^{(\alpha + i\beta)(X_t - X_0)}, \quad \alpha \neq 0,$$

with both stochastic radius and stochastic angle. Its induced Itô dynamics are

$$\frac{dZ_t}{Z_t} = \left[(\alpha + i\beta)\mu + \frac{1}{2}(\alpha + i\beta)^2\sigma^2 \right] dt + (\alpha + i\beta)\sigma dW_t,$$

and its exact random-walk step is

$$Z_{n+1} = Z_n \exp[(\alpha + i\beta)(\mu\Delta t + \sigma\Delta W_n)].$$

The radius reconstructs X_t ; the angle supplies a geometric rotation; and the shared Brownian increment forces rank-one local covariance. A genuine planar Brownian motion remains a different, rank-two, two-driver model. At scalar zero crossings, the ordinary polar logarithm must give way to localization and Tanaka local time.

The bridge between complex numbers and Itô calculus is therefore precise but limited: complex exponentials organize stochastic scale and rotation, while quadratic variation—not the algebraic identity $i^2 = -1$ —produces the Itô correction.

14 Appendix A. Expanded Itô algebra

Starting from Eq. 14 and Eq. 13,

$$\begin{aligned}
\frac{dZ}{Z} &= \alpha\mu dt + \alpha\sigma dW + i\beta\mu dt + i\beta\sigma dW \\
&\quad + \frac{1}{2}(\alpha^2 - \beta^2)\sigma^2 dt + i\alpha\beta\sigma^2 dt \\
&= \left[(\alpha + i\beta)\mu + \frac{1}{2}(\alpha^2 - \beta^2 + 2i\alpha\beta)\sigma^2 \right] dt \\
&\quad + (\alpha + i\beta)\sigma dW \\
&= \left(c\mu + \frac{1}{2}c^2\sigma^2 \right) dt + c\sigma dW.
\end{aligned}$$

The symbolic notebook independently simplifies the difference between the direct and polar drifts to zero.

15 Appendix B. Exact moment derivation

For any complex q , the entire Gaussian moment-generating function gives

$$\mathbb{E}[e^{qY_t}] = \exp(q\mu t + \frac{1}{2}q^2\sigma^2 t), \quad Y_t = X_t - X_0.$$

Taking $q = p\alpha$ and multiplying by R_0^p gives Eq. 18. Taking $q = nc$ gives Eq. 19. No complex branch choice appears because the random variable is defined by the entire exponential function, not by applying a logarithm to sampled Z_t .

16 Appendix C. Reproducibility manifest

The preserved execution used:

Table 9: Software used for the executed notebook and manuscript.

Component	Version
Python	3.12.13
NumPy	2.5.1
SciPy	1.18.0
pandas	3.0.3
Matplotlib	3.11.1
Seaborn	0.13.2
SymPy	1.14.0
nbformat	5.10.4
Quarto	1.9.38
TinyTeX	2026

From the repository root, regenerate and execute the notebook with:

```
conda run -n phase3-paper python \
  paper/notebooks/build_phase3_discovery.py

conda run -n phase3-paper jupyter nbconvert \
  --to notebook --execute --inplace \
  --ExecutePreprocessor.timeout=900 \
  paper/notebooks/phase3_discovery.ipynb
```

Then render the paper:

```
quarto render paper --to pdf
```

The notebook writes all figure sources to `paper/figures/`, all numerical tables to `paper/tables/`, and a machine-readable summary to `paper/tables/numerical_summary.json`.

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